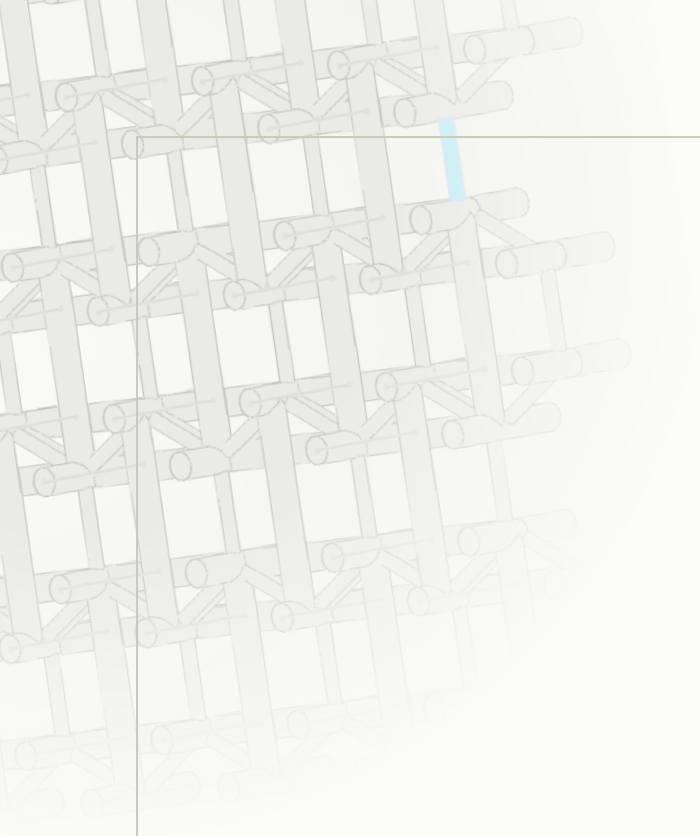
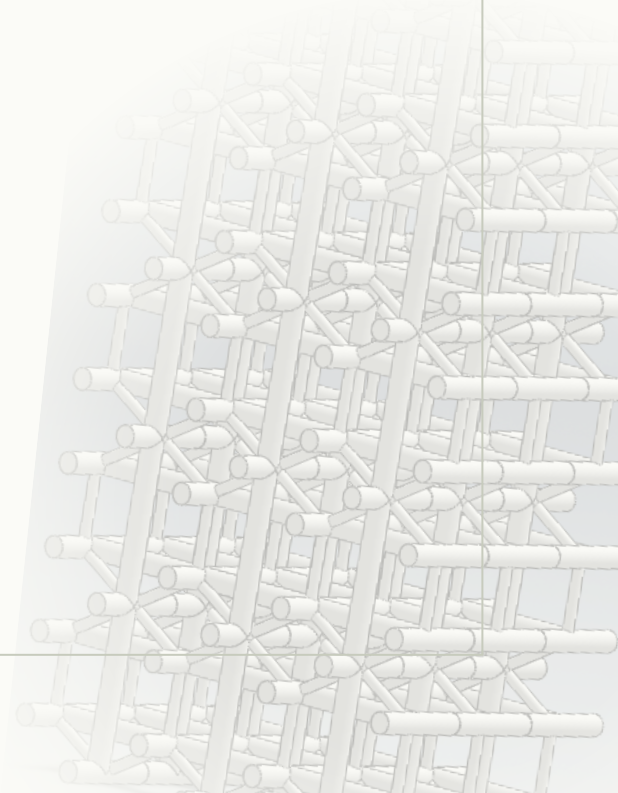
MECHANICS OF SOLIDS PROJECT

TENSEGRITY-INSPIRED LATTICES

STRUTS, CABLES AND GRADED HEXAGONAL CELLS

MOS PROJECT 2026–27 · TENSEGRITY STRAND





CONTENTS

1 Tensegrity, from structures to cells 3

1.1 What a tensegrity is 3

1.2 The hexagonal module and its continuum model 4

1.3 Reproducing the module properties 5

2 Lattices and experiments 7

2.1 Graded lattices 7

2.2 One test, analysed 8

2.3 The project's own lattice CAD 9

2.4 Observations and next steps 11

— References 12

Summary

A tensegrity is a network of compression struts held in place by tension cables. Gilewski et al. (2025) turned this idea into two-dimensional, 3D-printable hexagonal cells and derived closed-form elastic properties for them. This report re-implements that continuum model: it reproduces the paper's module properties to every printed digit, and maps how stiffness and Poisson's ratio depend on the cable-to-strut stiffness ratio k and the self-stress σ. The one rotated module in the family is the outlier — about three times softer vertically and ten times more strongly coupled sideways than the regular cells.

A raw compression record of the graded lattice with a rotated layer (M.hex.1.gra1) is analysed alongside: it peaks at 47.2 N, within 3 % of the paper's mean, then settles into a long plateau near 36 N as the soft layer shears. The project's own 2D and 3D lattice CAD is documented at the end.

PART ONE

1

TENSEGRITY, FROM STRUCTURES TO CELLS

Struts that never touch, held by cables that never slacken

1.1 What a tensegrity is

The word joins *tension* and *integrity*. A tensegrity is a pin-jointed system of compression members (struts) connected only through tension members (cables); the struts do not touch one another. It holds its shape not because its joints are rigid but because the cables are pre-tensioned: a self-equilibrated set of internal forces, the *self-stress*, stiffens mechanisms that would otherwise be free to move [1, 2]. Change the self-stress and the stiffness changes with it — which is what makes tensegrities attractive as tunable structures and, scaled down, as mechanical metamaterials.

Printing a true pin-jointed tensegrity is hard. Recent work therefore uses *tensegrity-inspired* lattices with rigid joints, which still show much of the characteristic tensegrity response [2]. Gilewski, Zawadzki, Thammavong and Al Sabouni-Zawadzka build two-dimensional lattices of hexagonal tensegrity modules, print them by MultiJet Printing, and test them in compression — the reference study for this strand [2].

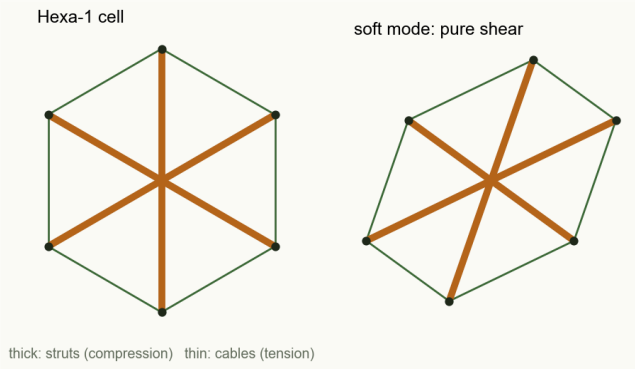


Fig. 1 A hexagonal tensegrity cell, schematic. Struts (amber) run between opposite nodes and pass each other without connecting; cables (green) close the hexagon. At the soft-mode limit the cell shears freely (right).

Struts carry compression and do not touch each other.

Cables carry tension only and set the shape.

Self-stress σ is the pre-tension; it stiffens the cell's mechanisms.

k is the cable-to-strut axial stiffness ratio.

1.2 The hexagonal module and its continuum model

The paper replaces each module by an equivalent 2D plane-stress continuum with an elasticity matrix E, derived from the axial stiffnesses of struts and cables and the self-stress [2]. Two parameters control everything:

$$k = \frac{(EA)_{\text{cable}}}{(EA)_{\text{strut}}}, \qquad \sigma = \frac{S_0}{EA} \tag{1}$$

where S_0 is the self-stress multiplier. For the regular Hexa-1 module the matrix is orthotropic, and its first eigenvalue

$$\lambda_1 = \frac{EA}{8ah}\left(\sqrt{3}\,k - 2\sigma\right) \tag{2}$$

vanishes on the line $\sigma_{\text{soft}} = \sqrt{3}\,k/2$. There the cell has a zero-energy deformation — pure shear for Hexa-1, a mixed compression–shear mode for the 60°-rotated Hexa-4. These are *extremal* materials: very stiff in some modes, almost free in one. The technical coefficients of Hexa-1 follow in closed form, with $N = 8\sqrt{3} + 9k^2 + 2(3+\sqrt{3})\sigma + k(9 + 9\sqrt{3} + 6\sqrt{3}\sigma)$:

$$E_1 = \frac{EA}{ah}\,\frac{N}{8\sqrt{3} + 9\sqrt{3}k + 6\sigma}, \qquad E_2 = \frac{EA}{3ah}\,\frac{N}{8 + 3\sqrt{3}k + 2\sigma} \tag{3}$$

$$G_{12} = \frac{EA}{8ah}\left(\sqrt{3}k - 2\sigma\right), \qquad \nu_{12} = \frac{3\sqrt{3}k - 6\sigma}{8\sqrt{3} + 9\sqrt{3}k + 6\sigma}, \qquad \nu_{21} = \frac{\sqrt{3}k - 2\sigma}{8 + 3\sqrt{3}k + 2\sigma} \tag{4}$$

Unlike the auxetic cells of the other strands, both Poisson's ratios here are positive — and for the regular cell very small. The shear modulus is the property that self-stress controls most directly: raising σ toward $\sqrt{3}k/2$ drives G_{12} to zero.

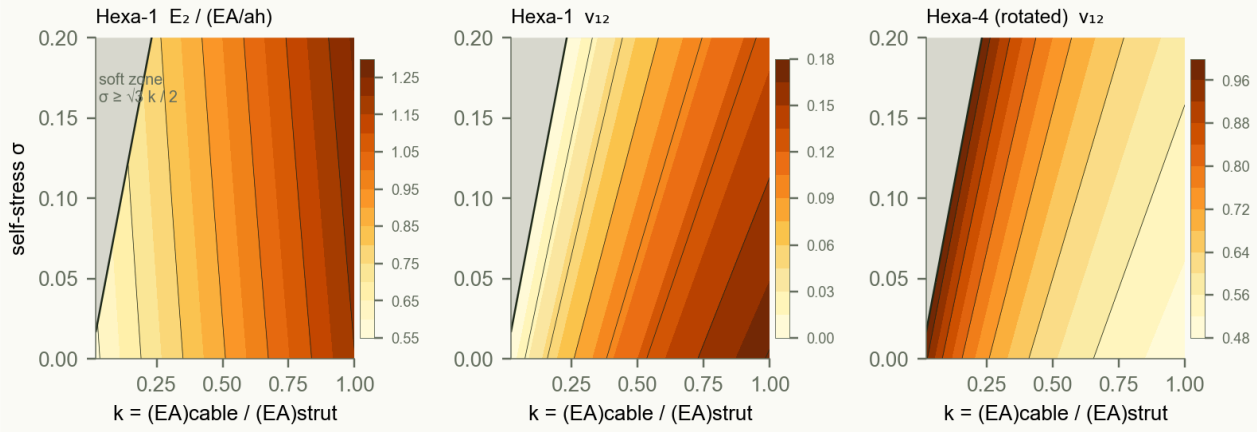


Fig. 2 Maps of the continuum model over 0 < k < 1 and 0 < σ < 0.2, computed from Eqs. (3)–(4) and the paper's Hexa-4 expressions. Grey: the soft zone σ ≥ √3k/2 where the elasticity matrix is not positive definite. The rotated module's Poisson's ratio is an order of magnitude above the regular one's and grows steeply toward the soft line.

1.3 Reproducing the module properties

The paper tabulates its printed module variants for a = 7.5 mm, cable diameter d_c = 0.75 mm and σ = 0.01, in units of a comparative modulus E. The formulas were re-implemented in Python, with $k = (d_c/d_s)^2$ for cables and struts of the same material and the equivalent thickness h taken as the strut diameter d_s — the choice that reproduces the published values.

Module	d_s (mm)	k	E_1 [×E]	E_2 [×E]	G_{12} [×E]	ν_{12}	ν_{21}
Regular-1a	1.5	0.250	0.1824	0.1162	0.0081	0.0696	0.0443
Regular-1b	2.0	0.141	0.2288	0.1404	0.0059	0.0416	0.0255
Regular-1c	2.5	0.090	0.2776	0.1670	0.0044	0.0266	0.0160
Rotated (Hexa-4)	1.5	0.250	0.0355	0.0376	0.0238	0.7357	0.7789

Table 1. Module properties recomputed from the closed-form model. Every value matches the paper's Table 1 to the digits printed there [2].

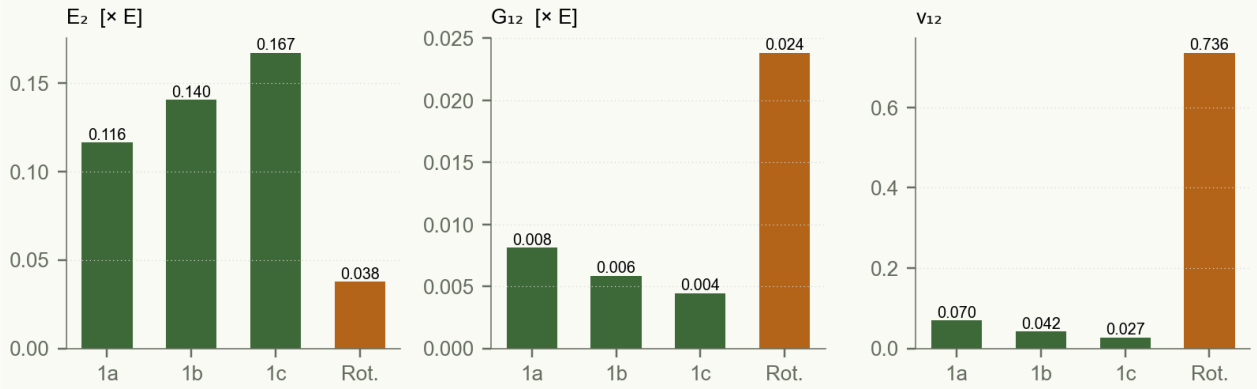


Fig. 3 Thicker struts (1a → 1c) stiffen the regular cell and make it slightly more shear-compliant. Rotating the same cell by 60° (amber) cuts E_2 to a third, triples the shear stiffness and raises the Poisson's ratio tenfold.

Two levers, opposite effects. Thickening the struts is a gentle, predictable dial: E_2 rises by 21 % and 44 % from 1a to 1b and 1c, while the Poisson's ratio falls. Rotating the module is a switch: the invariants of the elasticity tensor are unchanged, but the stiffness is redistributed into shear and lateral coupling, and the rotated cell gains non-zero shear–extension coupling terms that the regular cells lack. A layer of rotated cells is therefore a built-in weak, laterally active band — exactly what the graded lattices exploit.

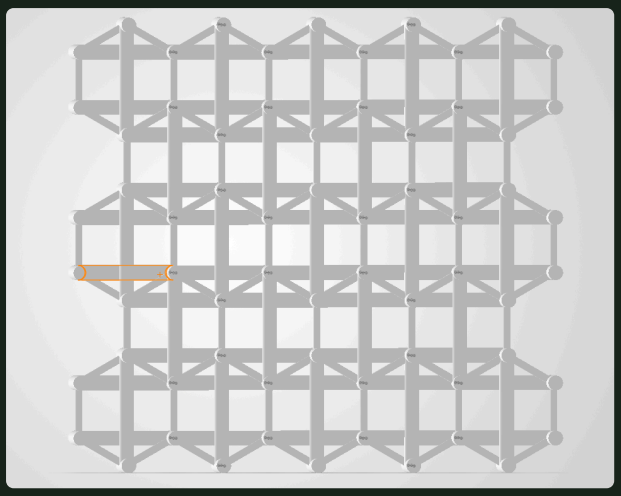


Fig. 4a Project CAD: a 2D hexagonal tensegrity-inspired lattice, front view.

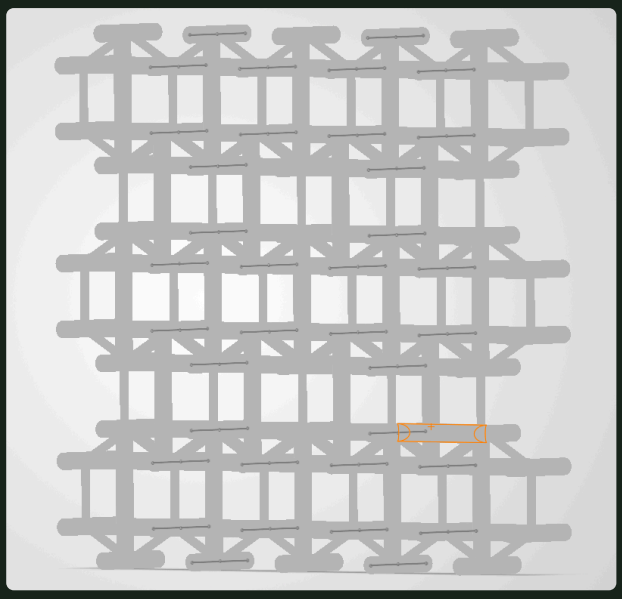


Fig. 4b The same lattice with its in-plane cables shown as thin lines between the joints.

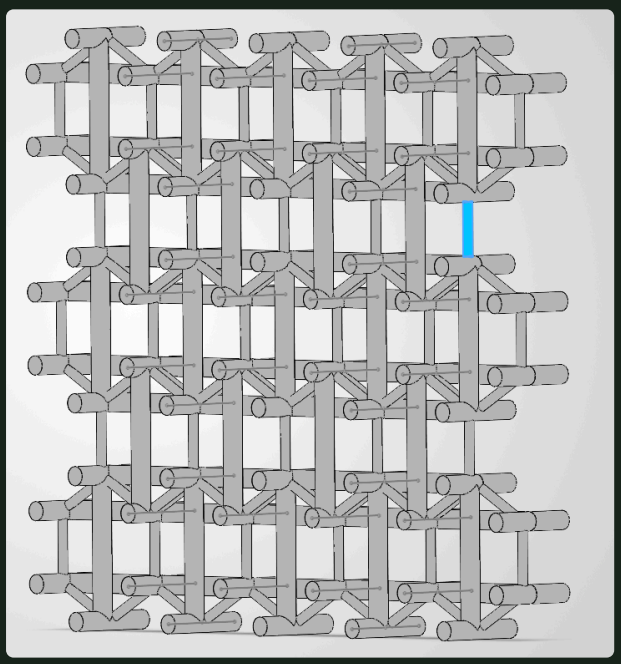


Fig. 4c Three-quarter view: struts and cables lie in two parallel planes, joined by short transverse stabilisers.

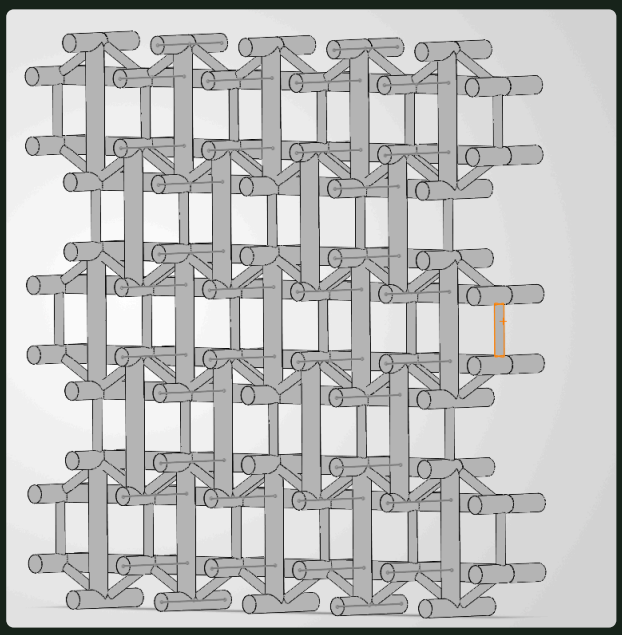


Fig. 4d The same view with one stabilising bar highlighted.

PART TWO

2

LATTICES AND EXPERIMENTS

Grading the cells to place the damage

2.1 Graded lattices

The reference study prints 10 × 6 lattices of these modules in VisiJet M2R-WT resin (tensile modulus 2000 MPa) and compresses them between greased glass plates at 5 mm/min, under either a uniform load or a load on the two middle top nodes [2]. Five layouts are tested: a regular lattice (uni1), one with a rotated layer (gra1), two with stiffened upper layers (gra2 by strut thickness, gra4 by module height), and one with a local, centric reinforcement (gra5).

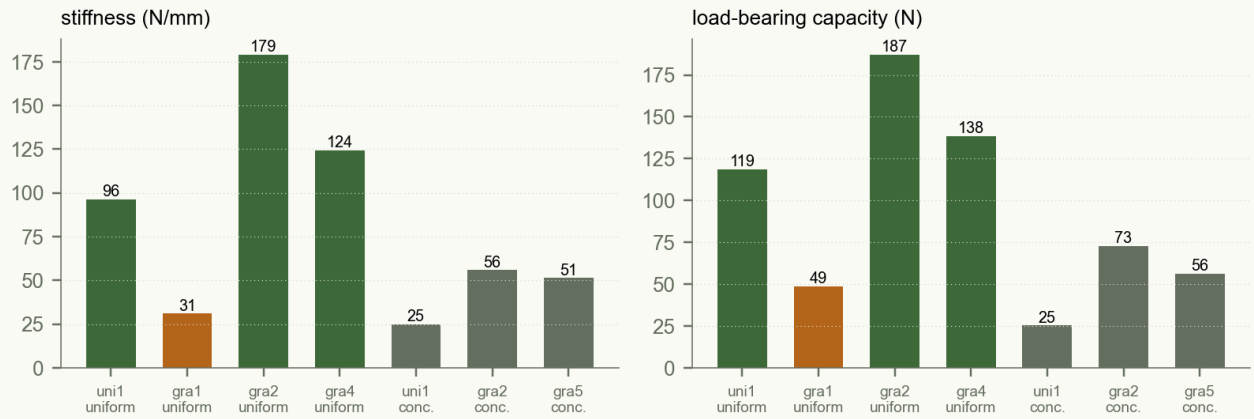


Fig. 5 Measured stiffness and load-bearing capacity of the tested lattices, from the paper's Table 3 [2]. Amber: the lattice with a rotated layer; grey: concentrated loading.

- **The weakest layer decides.** Under uniform load a lattice fails in its softest layer. The rotated layer of gra1 cuts stiffness to 32 % of the regular lattice — exactly the ratio of the two modules' E_2 in Table 1.
- **Stiffening the top moves the damage down.** gra2 and gra4 raise the capacity by 58 % and 17 % and push the failure into lower layers, leaving the reinforced top as a *mechanically stable region*.
- **Point loads spread when the top is stiff.** A concentrated load crushes two cells of the regular lattice; with stiffened upper layers the deformation delocalises over many cells and the capacity rises 2–3 times. Reinforcing just the eight cells near the load (gra5) gets almost the same gain as stiffening twenty.

2.2 One test, analysed

The folder holds the raw record of one specimen of the graded lattice with a rotated layer (M.hex.1.gra1), from the paper's open dataset: 1258 samples of time, crosshead displacement and force over 125 s, to 10.5 mm of travel.

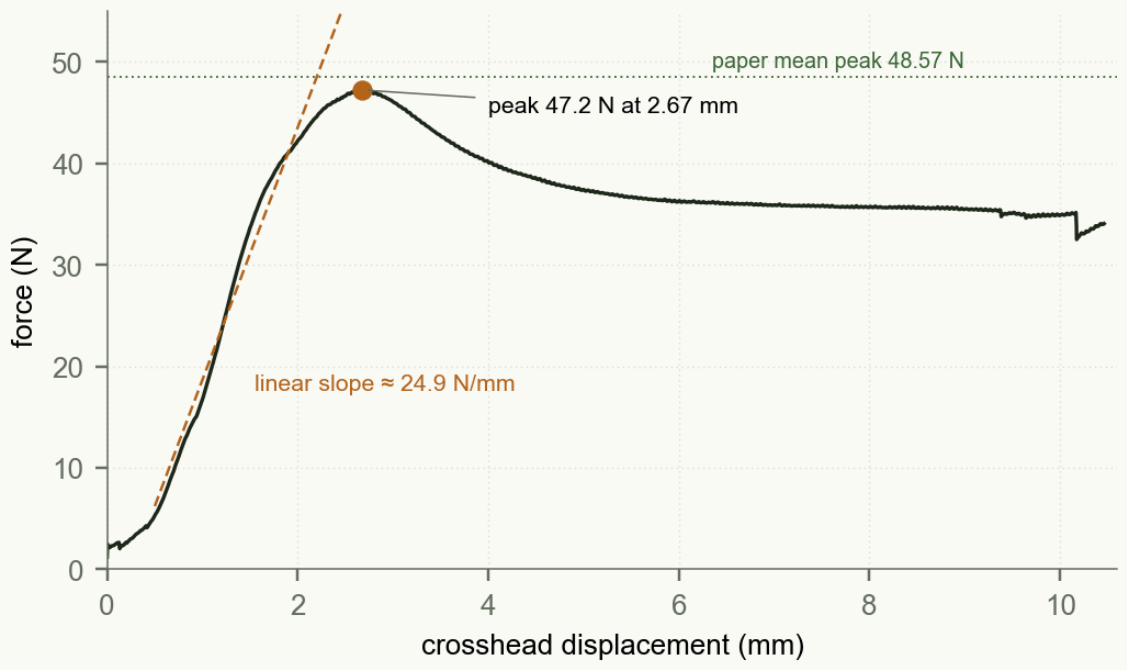


Fig. 6 Force against displacement for one M.hex.1.gra1 specimen. Dashed: straight-line fit over 0.8–2.2 mm.

Quantity	This specimen	Paper (mean of five) [2]
Peak force (N)	47.2	48.57
Displacement at peak (mm)	2.67	2.41
Linear stiffness (N/mm)	24.9	31.0
Post-peak plateau (N)	≈ 36	—

Table 2. Single-specimen values against the paper's averages.

The curve has three stages. A short seating toe up to about 0.5 mm; a near-linear rise of about 25 N/mm; and, past the 47 N peak, a soft decline into a long, flat plateau near 36 N that lasts to the end of the test. The peak sits within 3 % of the paper's mean. The fitted stiffness is about a fifth lower than the published 31.0 N/mm — expected for a single specimen and a hand-chosen fit window, against a value the paper derives from the mean of five curves.

The plateau is the signature of the rotated layer: once it starts to shear, the cells above and below stay almost undeformed and carry the load through a band that keeps deforming at nearly constant force. For energy absorption this is useful behaviour — a graded tensegrity lattice with a deliberate soft layer acts as a built-in load limiter.

2.3 The project's own lattice CAD

The strand's CAD models follow the reference lattice: hexagonal modules whose struts and cables sit in two parallel planes, tied by short transverse bars that keep the lattice flat during compression without carrying structural load [2]. Two forms were modelled — a planar 2D sheet (Fig. 4) and a three-dimensional column built by stacking and offsetting the same cells in depth (Fig. 7).

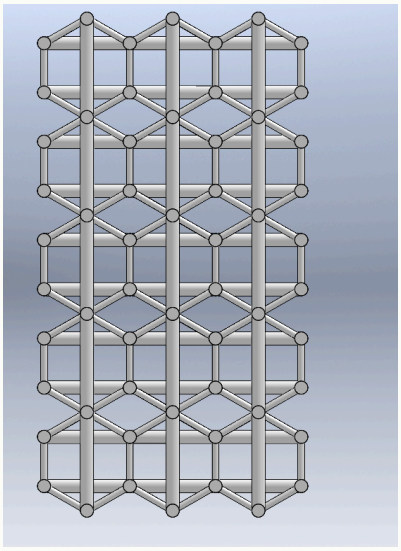


Fig. 7a 3D column, front view: the hexagonal pattern repeats through the depth.

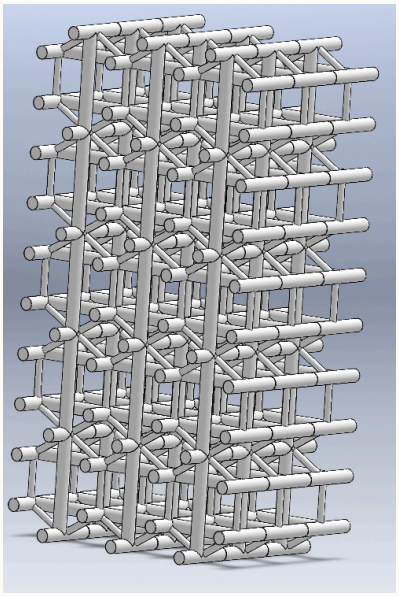


Fig. 7b Isometric view of the 3D column.

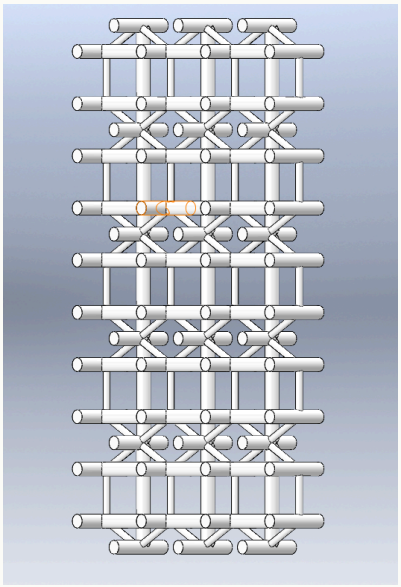


Fig. 7c Side view showing the stacked layers.

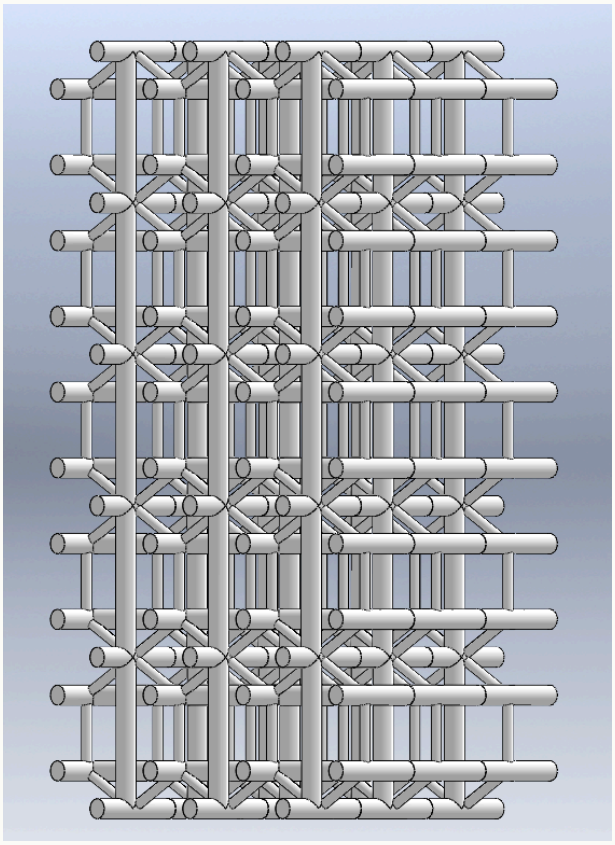


Fig. 7d 3D column, rear three-quarter view.

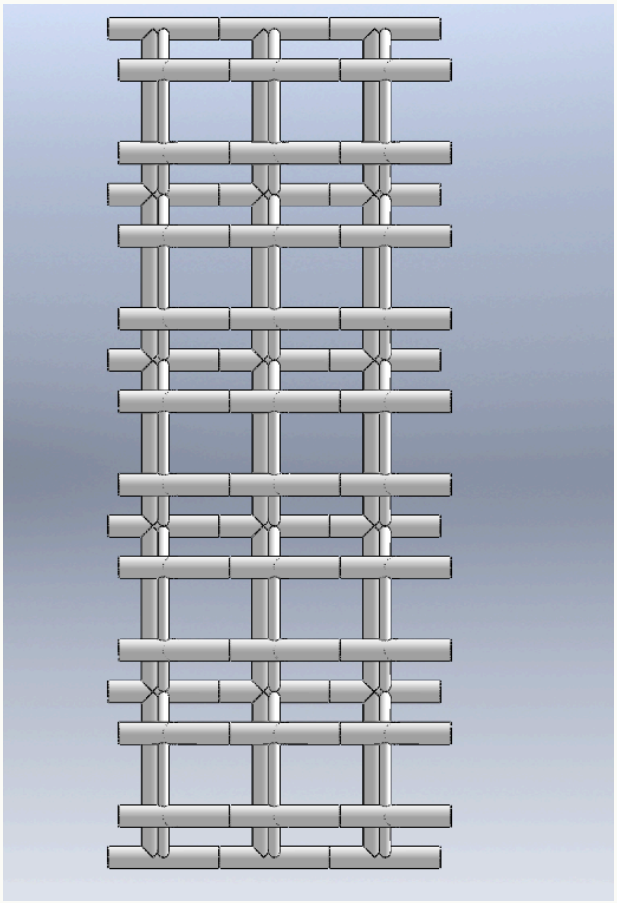


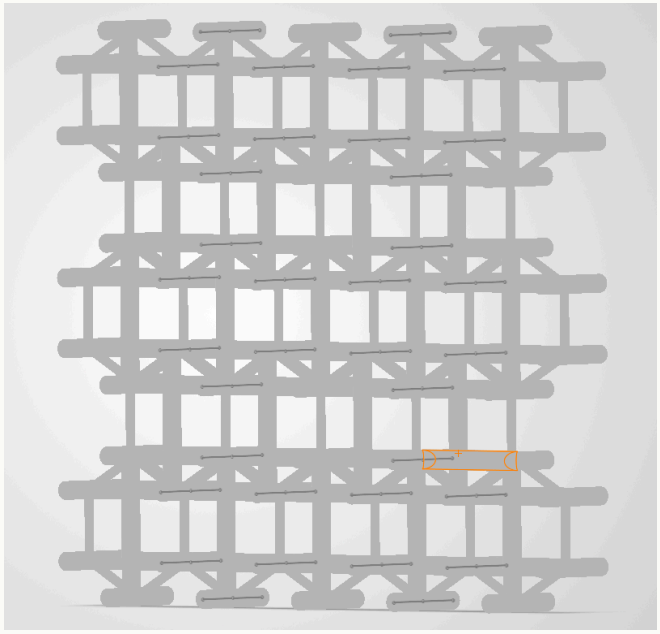
Fig. 7e 3D column, profile: the cell rows interlock through the depth.

Extending the 2D module into depth is the natural next step from the reference study, which stops at planar lattices. Two questions follow directly from Part 1. First, whether the soft modes survive in 3D: in the plane, rotating a module by 60° creates a shear-dominated soft band; in a stacked column, neighbouring layers may brace that band and remove it. Second, whether the transverse bars, harmless stabilisers in a single sheet, become load paths once several sheets are joined.

2.4 Observations and next steps

- **The continuum model is reproducible.** Implemented from the published formulas, it returns the paper's module properties exactly, so it can be used with confidence to explore new k, σ and geometry combinations before printing.
- **Orientation is the strongest design lever.** At the same material and member sizes, a 60° rotation changes E_2 three-fold and the Poisson's ratio ten-fold. Strut thickness and module height are finer dials.
- **Grading places the damage.** The tests show that the softest layer fails first and the stiffened layers stay intact, so a lattice can be designed to protect one region and sacrifice another.
- **The model is comparative, not predictive.** It describes pin-jointed cells, while printed lattices have rigid joints and the cell properties are scaled by an arbitrary E; the paper itself uses it only to rank and grade cells, not to predict forces [2].

Next steps for the strand. Print the project's 2D lattice and compare its curve with the M.hex.1 family; model the 3D column with the same continuum approach layer by layer; and test whether introducing self-stress in a printed lattice (for example by pre-tensioned cables) can move a cell along the k–σ maps of Fig. 2 rather than fixing it at print time.



REFERENCES

[1] Skelton, R. E., & de Oliveira, M. C. (2009). *Tensegrity Systems*. Springer.

[2] Gilewski, W., Zawadzki, A., Thammavong, M., & Al Sabouni-Zawadzka, A. (2025). 2D graded tensegrity-inspired metamaterials with hexagonal cells fabricated via additive manufacturing. *2D Materials*, 12, 045026. doi:10.1088/2053-1583/ae1620

[3] M_hex_1_gra1_RawData_1.csv — raw force–displacement record of one M.hex.1.gra1 specimen under uniform load, from the open dataset of [2].